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Adenosine and Its Derivatives Inhibit the cAMP Signaling Response in *Dictyostelium discoideum*

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In developmentally competent *Dictyostelium discoideum* amoebae, binding of cAMP to high-affinity surface receptors produces a rapid activation of adenylate cyclase which adapts within minutes. The result is a transient increase in intracellular cAMP which is rapidly secreted. Adenosine inhibited this cAMP signaling response with an apparent K_i of 300 μM. The apparent K_i's for 2′-*O*-methyladenosine and 2-chloroadenosine were approximately 30 and 100 μM, respectively. Inhibition by adenosine was rapid, reversible, and depended on the cAMP stimulus concentration. In addition, the adaptation of the cAMP signaling response was blocked by adenosine. As has been previously reported, adenosine inhibits cAMP binding to intact cells. Under the same developmental conditions as in the perfusion studies, we find the binding inhibition depends on both the cAMP and adenosine concentrations, with an apparent K_i of 100 μM. The apparent K_i's for 2′-*O*-methyl- and 2-chloroadenosine were approximately 8 and 35 μM, respectively. However, with cells developed for short times and with an axenic strain, inhibition was independent of cAMP concentration or cells showed mixed-type binding inhibition. The effect of adenosine on the cAMP signaling response is consistent with the reported effects of adenosine on other cAMP-mediated processes such as chemotaxis and the increase in intracellular cGMP, and may provide an explanation for the reported inhibition of center formation.
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INTRODUCTION

In *Dictyostelium discoideum* starvation of single amoebae triggers a developmental program which ultimately leads to the formation of two cell types—stalk and spore cells. Early in development, cell monolayers divide into territories of 10^5–10^6 cells and cells within the territory aggregate to the center. Aggregation is mediated by extracellular adenosine 3′:5′-cyclic monophosphate (cAMP). Periodically secreted by cells in the center of a territory, cAMP diffuses to nearby cells and binds to high-affinity surface receptors, leading to transient activation of adenylate cyclase. The result is a rise in intracellular cAMP which is rapidly secreted. Referred to as the cAMP signaling response, this process is repeated in more distal cells and produces a coordinated cell-to-cell relay of cAMP in the form of cAMP waves which propagate through the monolayer. These waves provide gradients which guide the chemotactically sensitive cells toward the central emitting source (Devreotes, 1982; Gerisch, 1982; Van Haastert and Konijn, 1982). In addition to its role in early development in signal relay and chemotaxis, cAMP has been implicated in the regulation of gene expression throughout development (Lodish *et al.*, 1982) and in differentiation of the multicellular organism through the slug stage to stalk and spore cells (Sussman, 1982).

Recently, Newell and Ross reported that adenosine, when added to the support agar or sprayed on developmentally competent cells could, in a dose-dependent manner, decrease the number and increase the size of territories in a cell monolayer (Newell and Ross, 1982). Bradley also reported a similar effect of adenine (Bradley *et al.*, 1956). The observation by Newell and Ross suggested that adenosine can block center formation but does not affect chemotaxis or signal relay. They reported that adenosine inhibits up to 90% of the binding of cAMP to cells, yet observed no inhibition of the other cAMP-mediated responses tested. However, Brenner has observed inhibition of the cAMP dependent rise in cAMP by adenosine (Brenner and Thoms, 1984). And in addition, studies recently reported by Van Haastert indicate that adenosine blocks chemotaxis to cAMP and inhibits the rise in cGMP as well as inhibits binding of cAMP to its receptor (Van Haastert, 1983). He concluded that these inhibitory effects were independent of test stimulus concentration, consistent with the observation of Newell and Ross that center formation was inhibited to the same extent by adenosine concentrations ranging from 5 to 20 mM and that the amount of binding inhibition did not depend on the cAMP concentration (Newell and Ross, 1982).

Using a perfusion assay to control the cAMP stimulus concentrations, we have examined the effect of adenosine on the cAMP signaling response. We have also measured the inhibition of cAMP binding by adenosine and the adenosine analogs 2-chloroadenosine and 2′-*O*-methyladenosine. We report here that adenosine and these adenosine analogs rapidly inhibit the cAMP

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dependent rise in cAMP. The amount of inhibition depends on both the concentration of adenosine and the concentration of cAMP stimulus. Consistent with this, we have found that adenosine and these analogs inhibit cAMP binding and under developmental conditions similar to those used in the perfusion experiments, the observed binding inhibition also depends on both the adenosine and cAMP concentrations. Under different developmental conditions, however, noncompetitive or mixed-type inhibition was observed. Since adenosine can inhibit cAMP binding to its receptor (either directly or indirectly through an adenosine receptor), it could inhibit processes requiring receptor occupancy. If the amount of inhibition by adenosine depends on the concentration of cAMP used to test the response, this may explain the discrepancy between the sensitivity of cAMP-mediated processes to adenosine between that shown here for cAMP signaling and that reported by Newell and Ross (1982) and Van Haastert (1983) for chemotaxis and cGMP accumulation. Furthermore, inhibition of the cAMP signaling response may explain the inhibitory effect of adenosine on center formation observed *in vivo*.

MATERIALS AND METHODS

Perfusion. NC-4 cells were grown as described (Devreotes *et al.*, 1979; Dinauer *et al.*, 1980a). NC-4 amoebae were washed off nutrient agar plates in DB (10 m*M* phosphate buffer (Na-K), pH 6.6, 2 m*M* $MgSO_4$, 200 μM $CaCl_2$) and freed of bacteria by four centrifugations (1500*g* for 4 min) at 4°C. Amoebae were shaken at 5×10^6 cells/ml (1.5 ml) with [^{3}H]adenosine-labeled *E. coli* (Devreotes *et al.*, 1979). After shaking for 3 hr at 22°C (100 rpm) amoebae were transferred to nutrient-free agar plates (35 mm VWR polystyrene) containing 3 ml of 2% agar in DB. After 30 min cells adhered and excess fluid was removed, lids were replaced, and cells were then incubated at 7°C for 17 hr in the dark. When the first signs of aggregation appeared, cells were gently washed off the plate with 2 ml of DB. Identical aliquots (200 μl) were pipetted onto eight Millipore filters (SSWP 01300) in the perfusion apparatus. Total radioactivity was between 2 and 6×10^6 cpm per filter. [^{3}H]cAMP was purified as described by Solomon (1979).

cAMP binding. The binding was assayed using the ammonium sulfate stabilization technique described by Van Haastert and Kien (1983). NC-4 amoebae were harvested off growth plates, washed with DB four times (1500*g* for 4 min) at 4°C to free them of bacteria, and resuspended at 2×10^7 cells/ml in DB. After 7–9 hr, cells were washed with 10 m*M* DTT in 10 m*M* phosphate buffer (Na-K), pH 6.6, and resuspended in this at 2×10^8 cells/ml. The final reaction volume was 100 μl and contained 10^7 cells. After binding [^{3}H]cAMP for 2 min at 0°C, 2 ml saturated ammonium sulfate at 0°C and 100 μl H_2O containing 1 mg of bovine serum albumin were added. The tubes were allowed to sit on ice for 5–10 min and then spun at 5000*g* for 10 min. The supernatant was aspirated and pellets were solubilized with 100 μl 1% SDS-10 m*M* Tris, pH 7.5, and counted with 3 ml of Liquiscint.

Analysis of inhibition data. In order to analyze the inhibition data simply, we assumed that adenosine inhibition depends only on the concentration of inhibitor and the K_i. We are of course aware of the fact that adenosine receptors have been reported on these cells (Newell, 1982, and Van Haastert, 1983). Van Haastert (1983) has proposed a model in which adenosine must first bind to an adenosine receptor and this complex inhibits cAMP binding to its receptor. However, the objective of this analysis was to determine only whether adenosine inhibition of signaling also depended on the stimulus concentration and obtain a quantitative measurement of inhibition. Models in which adenosine works alone or through a receptor could produce inhibition either dependent or independent of cAMP concentration. We are also aware that in the strictest sense, a K_i can only be defined for a system in which a single K_d can be defined. For signaling, we have assumed a single K_d where $K_d = K_s$, the concentration of cAMP that elicits a half-maximal response, even though this value was not required for the calculation of K_i. We have also assumed a single K_d for the binding data analysis. Inhibition data were plotted as the ratio of control to inhibited response as a function of adenosine concentration. A series of plots were generated which were linear and intercepted the y axis at 1. If the inhibition was independent of stimulus concentration, a single plot would have been generated. However, a series of plots were generated and the slope of each line was related to the cAMP stimulus concentration by the equations given in the figure legends. The inverses of these slopes were then replotted as a function of cAMP concentration to yield the plots shown in Fig. 2 and Fig. 5.

Materials. [^{3}H]Adenosine (20–50 Ci/mmol) and [^{3}H]-cAMP (27 Ci/mmol) were from ICN, Irvine, California. Adenosine, 2′-*O*-methyladenosine, 2-chloroadenosine, and cAMP were from Sigma, St. Louis, Missouri. Ammonium sulfate was from BRL, Bethesda, Maryland. All other chemicals and materials were from standard sources.

RESULTS

Inhibition of the cAMP Signaling Response

Typical cAMP signaling responses are shown in Figs. 1A and E. The rate of [^{3}H]cAMP secretion, which

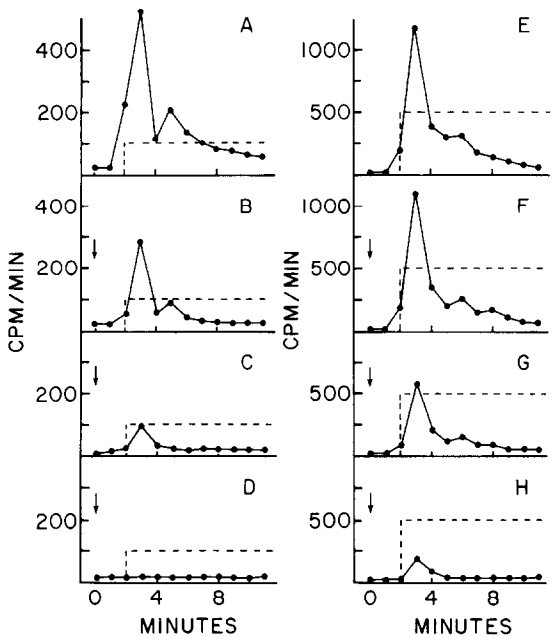

FIG. 1. Inhibition of cAMP signaling response by adenosine. Rates of [^{3}H]cAMP secretion were measured as described. Application of cAMP stimuli is denoted by the dashed rectangles. The start of addition of adenosine is shown by the arrow. The stimulus concentration in A–D is 10 nM cAMP. (A) Control. (B) 1 mM Adenosine. (C) 2.5 mM Adenosine. (D) 5 mM Adenosine. The stimulus concentration in E–H is 100 nM cAMP. (E) Control. (F) 5 mM Adenosine. (G) 10 mM Adenosine. (H) 20 mM Adenosine.

reflects the rate of synthesis of cAMP in response to a cAMP stimulus, rose and fell with a characteristic time course. As has been described (Devreotes *et al.*, 1979), cells respond to an increment in extracellular cAMP concentration, adapt within minutes to this concentration, and remain adapted as long as a constant stimulus is applied. The magnitude of the response depends on the concentration of cAMP stimulus (10 nM in 1A and 100 nM in 1E). To assess the effect of adenosine on the cAMP signaling response, adenosine was added 3 min before and during the addition of the cAMP stimulus. Figures 1B–D show the effects of adenosine at the lower stimulus concentration (10 nM). Addition of 1 and 2.5 mM adenosine reduced this response by 50 and 86%, respectively, and 5 mM completely blocked the response. Figures 1F–H show the inhibition by adenosine at a higher stimulus concentration (100 nM). Addition of 5 mM adenosine inhibited this response by only 5% (1F), 10 and 20 mM caused 50 and 90% inhibition. Note that adenosine had no effect on the time course of the response, but reduced the secretion rate to the same extent at each time point. In addition, inhibition was not due to a deleterious effect on cell viability or metabolism since the inhibition was rapidly reversible (data shown later).

These initial results indicated that the amount of inhibition depended on both the cAMP stimulus concentration and the adenosine concentration. We therefore carried out a more detailed study of this inhibition. The inhibition of the signaling response by various concentrations of adenosine and the adenosine analog 2′-*O*-methyladenosine was determined at many cAMP concentrations and analyzed as described under Materials and Methods. Figure 2A clearly shows the dependence of inhibition on stimulus concentration. A K_i for adenosine of 300 μM (Fig. 2A) was obtained. Since fewer experiments were performed with 2′-*O*-methyladenosine (Fig. 2B), fewer data points were obtained. To generate the curve, we drew the line to fit most of the points, excluding the highest concentration. This gives an approximate K_i of 30 μM, indicating that it is a much stronger inhibitor than adenosine and illustrating that the inhibition by *O*-methyladenosine also depends on the stimulus concentration. We also determined the K_i for the analog 2-chloroadenosine to be approximately 100 μM (data not shown). In addition, from the replots, the cAMP concentration that elicits a half-maximal response was observed to be between 2 and 6 nM.

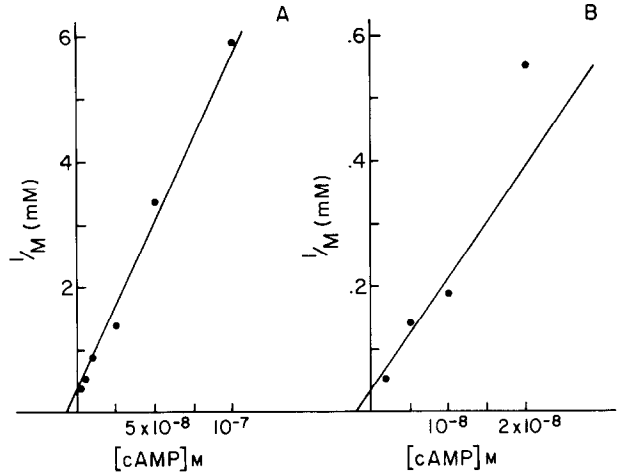

FIG. 2. Determination of K_i's for signaling inhibition for adenosine and 2′-*O*-methyladenosine. For this analysis, it was assumed that the dose–response data were approximated by the equation $V_0 = S/(K_s + S)$. The inhibited response V_i (total [^{3}H]cAMP released upon addition of cAMP in the presence of inhibitor) was compared to the uninhibited response, V_0 (total [^{3}H]cAMP released upon addition of cAMP in absence of inhibitor) and the ratio V_0/V_i was plotted as a function of adenosine and 2′-*O*-methyladenosine concentration. For competitive inhibition $V_0/V_i = 1 + K_sI/[K_i(K_s + S)]$ where I is the inhibitor concentration and K_i is the inhibitor concentration that gives 50% inhibition as $S \rightarrow 0$. This plot generates a series of lines that intercept the y axis at 1 and the slope of each line depends on the stimulus concentration [S]. Shown here are the inverses of the slopes of these lines ($1/M$) vs the stimulus concentrations [S] which are related by the equation $1/M = K_i + SK_i/K_s$. Therefore, the y intercept gives the K_i for the inhibitor and the x intercept is $-K_s$ for cAMP. (A) Adenosine. (B) 2′-*O*-Methyladenosine.

Rapid Kinetics of Inhibition

In these experiments, cells were pretreated with adenosine for 3 min before addition of the stimulus. The same amount of inhibition was observed whether the cells were pretreated with adenosine for 10 min or if it was added at the same time as the cAMP stimulus. However, a small delay in the onset of inhibition at the beginning of the response may not be detected when the entire response is measured since the cellular secretion rate requires between 1 and 2 min to reach a peak. To determine if there was a delay, the effect of the time of pretreatment with adenosine on the response to a brief application of the cAMP stimulus was examined. The control response to 50 n*M* cAMP for 30 sec is shown in Fig. 3A. Adenosine (20 m*M*) completely inhibited the response whether it was added at the same time as the cAMP stimulus (3B) or if it was added 30 sec (3C) or 60 sec (3D) before the stimulus. This result indicates that no pretreatment is required for inhibition and adenosine was effective within a 30-sec time period. To further establish the rapidity of adenosine inhibition, we tested whether addition of adenosine had the same effect as removal of the stimulus, which leads to an abrupt decline in the secretion rate (Dinauer *et al.*, 1980a). As shown in Fig. 3E, when after 2 min the cAMP stimulus (10 n*M* cAMP) was withdrawn, the cAMP secretion rate rapidly declined. When 10 m*M* adenosine was added after 2 min (3E), the cAMP secretion rate fell to zero with approximately the same kinetics as it did when the stimulus was removed. In Figs. 3F and G, this experiment was repeated except that the stimulus was interrupted (by addition of adenosine or removal of stimulus) at 1.5 min (3F) and 1 min (3G). These responses fell off with the same kinetics whether cAMP was removed or adenosine was added.

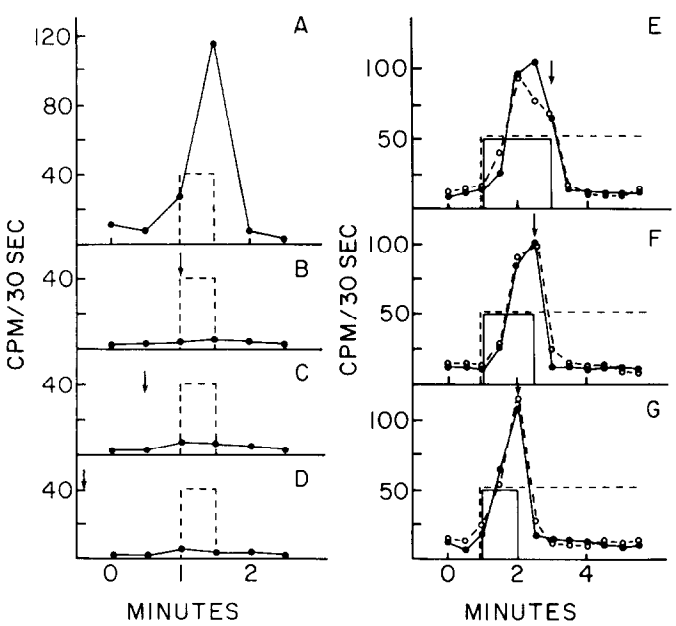

FIG. 3. Determination of the rapidity of adenosine inhibition. Addition of adenosine is denoted by the arrow. The stimulus concentration in A–D is 50 n*M* cAMP and is applied for 30 sec (shown by the dashed rectangle). Adenosine is 20 m*M*. In E–G, the cAMP stimulus (shown by the dashed and solid rectangles) is 10 n*M* and adenosine is 10 m*M*. The dashed curves show the response with the addition of adenosine shown by the arrow at (E) 2 min, (F) $1\frac{1}{2}$ min, (G) 1 min. The solid curves show the response if the stimulus is removed at (E) 2 min, (F) $1\frac{1}{2}$ min, (G) 1 min.

Inhibition of [^{3}H]cAMP Binding

The rapid inhibition of the cAMP signaling response by adenosine suggested that it may act by preventing cAMP binding to its surface receptor. Others have observed inhibition of cAMP binding by adenosine (Newell and Ross, 1982; Van Haastert, 1983). Although they reported the inhibition to be noncompetitive, it is clear from the physiological studies shown above that adenosine inhibition of the signaling response depended on the concentration of cAMP used to stimulate. We therefore reexamined the effect of adenosine on cAMP binding on cells developed as described under Materials and Methods (shaking at 2×10^7 cells/ml for 7–9 hr). In Fig. 4A, the binding data are plotted by the method of Scatchard (1949) and form a typical curvilinear plot that is characteristic of [^{3}H]cAMP binding to intact cells. Also shown in Fig. 4A is the cAMP binding in the presence of 150 μM adenosine and 15 μM *O*-methyladenosine. The inhibitors decrease the slopes of the binding curves while the x intercepts are unaffected, indicating that the amount of inhibition by adenosine or *O*-methyladenosine was dependent on the concentration of cAMP in the binding assay. To show this dependence more clearly, binding was assayed at either 100 or 1 n*M* [^{3}H]cAMP (Fig. 4B), and competed with either cAMP, adenosine or *O*-methyladenosine. The degree of inhibition of 1 n*M* cAMP is greater than at 100 n*M* cAMP for every competitor concentration tested. The binding data from the above experiments and others were combined, analyzed similar to the signaling inhibition, and are presented in Fig. 5. As before, we assumed a single K_d and from these plots a K_i for binding inhibition of 100 μM for adenosine and 8 μM *O*-methyladenosine was obtained. A K_i for 2-chloroadenosine was also determined (data not shown) to be 30 μM. A K_d for the cAMP receptor of 30–40 n*M* was observed. To try to discover the reason for the discrepancy between this result and previous reports, we tested binding inhibition on axenic cells and NC-4 cells developed for short times (4–5 hr). We clearly

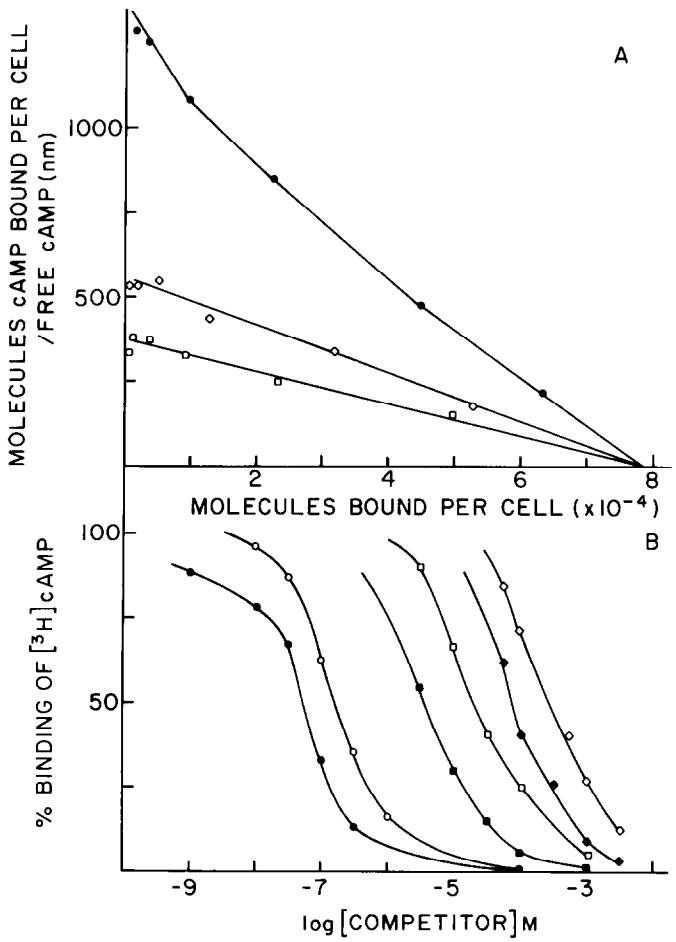

FIG. 4. Binding inhibition by adenosine and 2′-*O*-methyladenosine. (A) Scatchard plots of the binding of [^{3}H]cAMP to cell surface receptors in the absence (●) or presence of 150 μM adenosine (◇) or 15 μM 2′-*O*-methyladenosine (□). (B) Binding of 1 n*M* [^{3}H]cAMP (●, ■, ♦) or 100 n*M* [^{3}H]cAMP (○, □, ◇) was measured in the presence of increasing concentrations of competitor. cAMP (●, ○) 2′-*O*-methyladenosine (■, □) or adenosine (♦, ◇). Binding of 1 or 100 n*M* [^{3}H]cAMP in the absence of competitor was set at 100% and the binding in the presence of 100 μM cAMP was set at 0%.

observed noncompetitive or mixed-type binding inhibition under these conditions, a result which is similar to that shown by Newell and Ross (1982) and Van Haastert (1983).

Adenosine Prevents Adaptation

It has been previously proposed that both the process leading to activation of adenylate cyclase (excitation) and that leading to the deactivation of adenylate cyclase (adaptation) are directly controlled by receptor occupancy (Dinauer *et al.*, 1980b). Thus both of these responses should be inhibited by adenosine, if it blocks binding of cAMP to its receptor. An experiment designed to test this prediction is shown in Fig. 6. Figure 6A shows the control response to 10 n*M* cAMP. Figure 6B shows that pretreatment with 10 m*M* adenosine for 9 min had little effect on the cellular response, indicating that the effect of the drug is rapidly reversible. As shown in Fig. 6C, when adenosine is added with the stimulus, the response is blocked. The adenosine was withdrawn 7 min after the stimulus was added and replaced by the cAMP alone. As can be seen, the cells responded normally when the inhibitor was removed. These results demonstrate that adaptation as well as excitation is inhibited by adenosine since had adaptation proceeded in the presence of adenosine, no response would have been observed.

DISCUSSION

It has been demonstrated here that adenosine inhibits the cAMP signaling response (Fig. 1). Analysis of the inhibition data (Fig. 2) reveals that the degree of

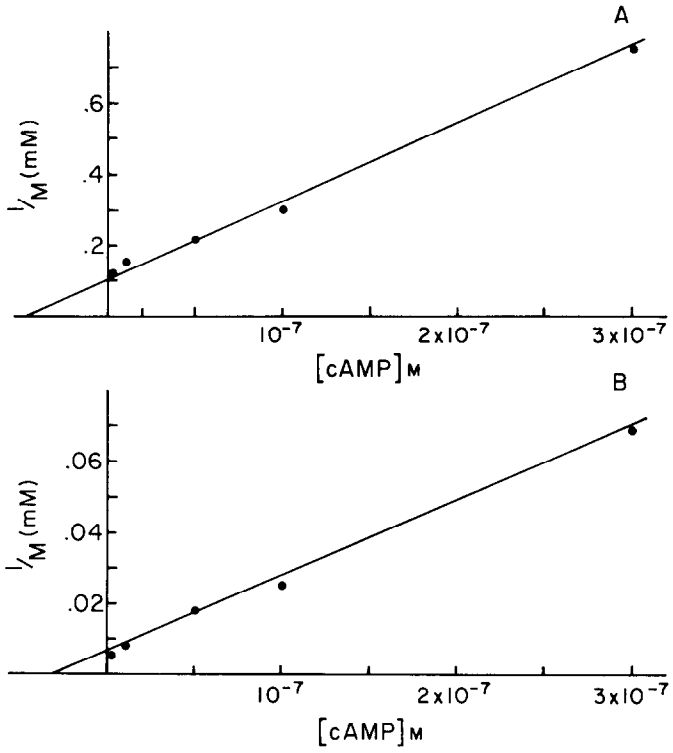

FIG. 5. Determination of K_i's for binding inhibition for adenosine and 2′-*O*-methyladenosine. The inhibition data were analyzed here as was described in Fig. 2. Here $B_0/B_i = 1 + K_d I/[K_i(K_d + S)]$ where B_0 is the amount of [^{3}H]cAMP bound in the absence of inhibitor, B_i is the amount of [^{3}H]cAMP bound in the presence of inhibitor, K_d is the dissociation constant of the cAMP receptor, I is the inhibitor concentration, K_i is the concentration of inhibitor that produces 50% inhibition as $S \rightarrow 0$, and S is the concentration of [^{3}H]cAMP used in the binding assay. The inverse of the slopes of these lines ($1/M$) was plotted vs the cAMP concentration in the binding. $1/M = K_i + S/K_dK_i$ to determine the K_i for (A) adenosine, (B) 2′-*O*-methyladenosine.

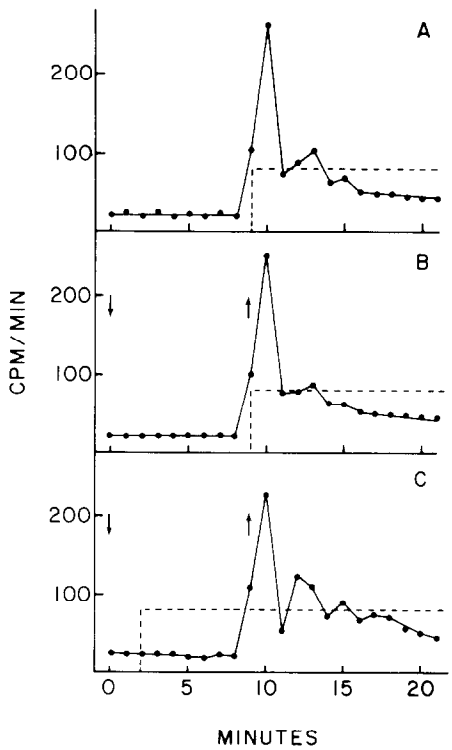

FIG. 6. Reversibility of adenosine action and effect of adenosine on adaptation. Addition and removal of 10 m*M* adenosine is shown by the arrows. The stimulus, shown by the dashed rectangle, is 10 n*M* cAMP. (A) The response to the stimulus. (B) The response to the stimulus after pretreatment with adenosine for 9 min. (C) The response to the stimulus after pretreatment with adenosine for 2 min then adenosine and cAMP for 7 min.

inhibition depends on both the adenosine and the cAMP stimulus concentration. 2′-*O*-methyladenosine is a more effective inhibitor of cAMP signaling (K_i = 30 μM) than is adenosine (K_i = 300 μM). The inhibition by adenosine was shown to be rapid. Less than 30 sec was required to inhibit the response (Fig. 3A) and when adenosine was added during a response, it led to a decline in the cAMP secretion rate with kinetics comparable to that caused by removal of the stimulus (Fig. 3B). The inhibition was also rapidly reversible since cells that had been pretreated with adenosine responded normally when adenosine was withdrawn and stimulus was added (Fig. 6B). This suggests that the effect of adenosine is not due to a loss of cell viability or a change in the ATP substrate pool. In addition, cells responded normally to the cAMP stimulus after pretreatment with both adenosine and cAMP (Fig. 6C), indicating that adenosine blocks not only the process leading to activation of adenylate cyclase but also that leading to adaptation.

These observations are in agreement with previously reported observations of the inhibitory effect of adenosine on the cAMP accumulation in suspension, chemotaxis, cGMP accumulation, and center formation. First, this report of signaling inhibition measured with the perfusion technique agrees with that reported by Brenner and Thoms (1984) in which they observed inhibition by adenosine of the rise in total cAMP in stimulated cells in suspension. Second, Van Haastert reported inhibition of both chemotaxis and the rise in cGMP under conditions of lower test stimulus concentrations (Van Haastert, 1983). However, he observed that attenuation of the cGMP response required pretreatment with adenosine before addition of the cAMP stimulus, whereas we found that no pretreatment was required for signaling inhibition (Fig. 3). A possible explanation is that since the cGMP response is so rapid (peak 10–20 sec) the equilibration time of adenosine binding may be a factor in the inhibition of this response.

Newell and Ross observed that adenosine changed the number and size of territories of aggregating cells, an effect they concluded was due to inhibition of center formation (Newell and Ross, 1982). The cAMP signaling response is a likely component involved in the formation of centers according to theoretical models that account for pulsatile release of cAMP in terms of feedback loops whose excitability can be raised to the point of autonomous generation of cAMP pulses. A prediction of these models is that a decrease in sensitivity of the signaling response (signal relay response) could change the excitability (but only slightly the ability to relay) of the system and thus eliminate spontaneous pulsing (Goldbeter and Segal, 1977). Our results show that adenosine does inhibit cAMP signaling, probably to a small degree at physiological cAMP wave concentrations (0.1 to 1 μM) and 5–10 m*M* adenosine. This slight inhibition may be sufficient to change the excitability of the system. Under the assay conditions used by Newell and Ross (1982, saturating cAMP test stimuli), they observed no decrease in chemotaxis or the rise in cGMP and also no change in the rate of wave propagation. If adenosine inhibition does depend on cAMP concentration, little inhibition would be expected at those concentrations of cAMP and adenosine. Indeed, we observed only 50% inhibition of the signaling response by 10 m*M* adenosine when a stimulus of 100 n*M* cAMP was applied (Fig. 1).

Consistent with previously reported results, we have shown here that adenosine inhibits binding of [^{3}H]cAMP to intact cells. In agreement with the results presented above for the signaling inhibition, the binding inhibition for cells developed similarly also depended on both the adenosine and cAMP concentrations (Fig. 4). A K_i for 2′-*O*-methyladenosine of 8 μM and for adenosine of 100

μM was obtained. Both Newell and Ross (1982) and Van Haastert (1983) showed that adenosine inhibited cAMP binding to cells but both of their results indicate that this inhibition was noncompetitive. A possible explanation of the discrepancy between their results and ours may lie in the fact that we observed a mixture of apparently competitive and noncompetitive inhibition in cells that were developed for shorter times and in an axenically grown strain.

Adenosine receptors have been reported in *D. discoideum* by Newell (1982) and Van Haastert (1983). Van Haastert has proposed that adenosine first binds to the adenosine β receptor and the adenosine/β receptor complex inhibits the binding of cAMP to the cAMP receptor. This model is consistent with the observations that the K_i for binding inhibition is similar to the K_d of the β receptor (350 μM) and that 2′-*O*-methyladenosine is a more effective inhibitor of signaling and binding. Both competitive and noncompetitive inhibition could be produced in this model. Either type could also be produced by direct antagonism of the receptor. Since either type of inhibition could be observed in either model, it is not clear whether the mechanism of action of adenosine is direct or via an adenosine receptor.

Is there a physiological role for adenosine during *Dictyostelium* development? Brackenbury reported that addition of the adenosine analog formycin B to migrating slugs could change the morphogenetic pathway (Brackenbury *et al.*, 1974). In addition, Armant showed the presence of 5′-nucleotidase in prestalk cells at the junction between prestalk and prespore cells (Armant *et al.*, 1980). In higher eukaryotes adenine nucleosides and nucleotides regulate numerous physiological processes via purinergic receptors (Burnstock and Brown, 1981). Adenosine and its analogs display prominent neurophysiological actions (Phillis and Wu, 1981) and have both stimulatory and inhibitory effects on adenylate cyclase (Cooper *et al.*, 1981; van Calker *et al.*, 1979). Adenosine has also been shown to inhibit platelet aggregation (Haslam and Cusack, 1981), dilate coronary blood vessels (Schrader, 1981), and have locomotor (Snyder *et al.*, 1981) and cardiac depressant effects (Schrader, 1981).

In *D. discoideum*, adenosine or its analogs may be useful in studies of the cAMP receptor. Are all cAMP-dependent processes inhibited by adenosine with a similar K_i? The answer to this question will help to establish whether a single receptor mediates all cAMP-dependent processes. In addition, adenosine may aid in identification of the cAMP receptor since it should be a specific competitor. Finally, adenosine may also be useful in determining the requirement for cAMP in early development, in gene expression throughout development, and in morphogenesis and differentiation.

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